

Project Proposal: Machine Learning with Graphs

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TL;DR. We represent a frozen ViT’s attention as a graph and train a GNN to predict failures beyond confidence and non-graph baselines.

1 Background & Motivation

Vision Transformers (ViTs) represent an image as patch tokens and repeatedly exchange information through multi-head self-attention [Dosovitskiy et al. 2021]. Reliability is usually estimated from the final logits, although predictions with similar confidence may arise from different layer-wise interactions. We therefore ask:

Can a GNN operating on a ViT’s multi-layer attention graph predict classification errors, especially under unseen corruptions, better than output-, representation-, and non-graph attention baselines?

This is useful for selective prediction, where likely failures are rejected or referred for review, and scientifically useful for testing whether errors exhibit recurring routing signatures. Each image naturally induces a directed, weighted graph over patch tokens, while correctness is a graph-level property of the frozen classifier.

Related work and gap. ViT attention has already been interpreted as a graph, but mainly for other objectives: relevance propagation for explanation [Chefer et al. 2021], weighted-PageRank token pruning [Wang et al. 2024], and graph regularization of class-patch interactions for weakly supervised segmentation [Yang et al. 2025]. Cultrera et al. [2023] use a convolutional autoencoder over ViT attention heatmaps for OOD detection, without learning on token-to-token connectivity. Closest in task, LogitDynamics predicts ViT errors from layer-wise class-logit trajectories, but does not use attention graphs [Beigelman and Freiman 2026].

NLP provides the closest methodological precedents. Handcrafted topological or spectral summaries of attention graphs have been used for generated-text and hallucination detection [Binkowski et al. 2025; Kushnareva et al. 2021], while HalluZig models how attention-graph topology evolves across layers [Samaga et al. 2026]. More directly, HaluGNN uses hidden states as node features and attention values as weighted edges [Kong et al. 2026b]. CHARM represents raw attention scores and activations as attributes of a token graph and applies edge-aware message passing; its ablations show a benefit from graph connectivity over a set model [Frasca et al. 2026]. CausalGaze further refines attention edges using gradient sensitivity before GNN classification [Kong et al. 2026a]. Thus, learned attention-graph diagnostics are established for language models. However, our literature search found no analogous method trained on per-image, multi-layer ViT patch-attention graphs to predict visual classification failure. We investigate this vision-specific gap.

2 Proposed Approach

Let $A^{(\ell,h)} \in \mathbb{R}^{N \times N}$ be the attention matrix at ViT layer ℓ and head h , including the class token. Since query token i reads information from token j , we orient information flow as $j \rightarrow i$. Following CHARM [Frasca et al. 2026], the raw per-head attention scores are used directly as edge features:

$$\mathbf{e}_{ji}^{(\ell)} = \left[A_{ij}^{(\ell,1)}, \dots, A_{ij}^{(\ell,H)} \right].$$

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To obtain a sparse graph without discarding interactions that are strong in only one specialized head, we retain the edge $j \rightarrow i$ whenever

$$\max_h A_{ij}^{(\ell, h)} > \tau.$$

Thus, the threshold determines only whether the edge is present; once retained, its edge feature contains the original attention scores from all heads. The threshold τ is selected on validation data.

Diagonal scores are stored as node features, together with normalized patch coordinates and a class-token indicator. We compare this attention-focused representation with a richer variant that also appends the corresponding ViT token embedding $\mathbf{z}_i^{(\ell)}$.

An edge-aware message-passing GNN uses $\mathbf{e}_{ji}^{(\ell)}$ when computing messages. We construct one attention graph for every ViT layer,

$$G^{(1)}(x), \dots, G^{(L)}(x),$$

and encode all layer graphs with a shared GNN:

$$\mathbf{g}_\ell = \text{GNN}\left(G^{(\ell)}(x)\right).$$

The resulting ordered sequence of graph embeddings

$$\mathbf{g}_1, \dots, \mathbf{g}_L$$

is then combined and followed by an MLP that predicts classification failure. Using all layers allows the detector to capture the complete evolution of attention routing, including early local interactions and later global aggregation. As ablations, we will compare this full layer-wise model with a last-layer-only model and a final-four-layers model.

For image x with label $y(x)$ and frozen ViT f , the automatic target is

$$y_{\text{err}}(x) = 1 \left[\arg \max_c f_c(x) \neq y(x) \right].$$

Only the detector is trained, using class-weighted binary cross-entropy. We compare:

- (1) the attention graph with minimal node features;
- (2) the attention graph augmented with token embeddings.

Novelty. The novelty is not the attention-as-graph view or GNNs over Transformer traces in general. It is their controlled evaluation for per-image ViT patch graphs, with comparisons against output signals, hidden representations, and non-graph models given the same attention data.

3 Experimental Plan

Datasets and setup. The primary benchmark will use CIFAR-100 and CIFAR-100-C, whose corruption families and severity levels follow Hendrycks and Dietterich [2019]. We will begin with a publicly available ViT checkpoint fine-tuned on CIFAR-100 and verify its accuracy, preprocessing pipeline, label mapping, confidence distribution, and access to attention matrices from all Transformer layers. If no suitable checkpoint is available, we will fine-tune an ImageNet-pretrained ViT on CIFAR-100. The selected ViT will then be frozen throughout all detector experiments.

If no publicly available checkpoint satisfies these requirements, we will fine-tune an ImageNet-pretrained ViT on the clean CIFAR-100 training set and select the checkpoint according to validation accuracy. In either case, the selected ViT will then be frozen and used as the fixed classifier throughout all detector experiments.

We will evaluate the selected classifier on CIFAR-100-C and report accuracy separately for each corruption type and severity level. After validating the classifier, we will precompute its attention graphs. The detector will be trained on clean images and a selected subset of noise, blur, weather, and digital corruptions, while complete corruption families will be reserved for testing. All corrupted versions of the same source image will remain in the same split to prevent leakage. Optional far-OOD datasets are SVHN [Netzer et al. 2011] and DTD [Cimpoi et al. 2014]; localized patches and occlusions will be generated automatically.

Baselines and ablations. We first report the frozen ViT’s clean and corruption-specific classification accuracy. For failure prediction, we compare against maximum softmax probability, logit margin, and an MLP on the final class-token embedding.

To isolate the contribution of graph structure, we also compare against an MLP receiving the same flattened attention tensors as the GNN and a sequence model receiving class-token embeddings from all ViT layers.

Graph ablations include last-layer-only input, the final four layers, all layers, and threshold sensitivity.

Metrics and success criteria. We evaluate error prediction using AUROC and AUPRC, treating an incorrect ViT prediction as the positive class. Since the detector may be used to reject unreliable predictions, we also report risk-coverage curves, area under the risk-coverage curve, and selective accuracy at fixed coverage levels.

Metrics are reported separately for clean images, corruption families seen during detector training, and unseen corruption families. The main hypothesis is supported if the GNN outperforms non-graph models given the same attention values, particularly on unseen corruptions.

Compute resources. The project requires GPU access provided by the course or university. The main compute cost is fine-tuning the ViT and precomputing attention graphs, token features, and logits for clean and corrupted images, as well as training the GNN over them.

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